

HURON COUNTY AP, MI, USA

WMO: 725406

Lat: 43.780N

Lon: 82.986W

Elev: 766

StdP: 14.29

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.3	(c) 2.6	(d) -8.5	(e) 3.6	(f) -2.0	(g) -3.1	(h) 4.8	(i) 4.4	(j) 25.6	(k) 24.9	(l) 23.7	(m) 23.6	(n) 5.6	(o) 260	(p) 0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	87.6	72.1	84.1	70.7	81.6	69.6	75.3	83.1	73.6	81.1	71.9	78.8	10.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.8	125.2	79.3	71.7	120.6	78.2	69.9	113.1	76.5	39.2	83.5	37.6	80.8	36.1	78.3	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	20.0	18.0	-10.9	91.5	8.1	2.4	-16.8	93.3	-21.5	94.7	-26.1	96.1	-32.0	97.9
			-11.3	78.7	8.1	1.8	-17.1	80.0	-21.8	81.0	-26.4	82.0	-32.3	83.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	21.7	22.4	31.7	43.5	54.9	64.0	68.1	66.6	60.5	49.2	38.3	28.0		
	DBStd	18.71	10.99	10.75	11.03	9.17	8.96	7.29	6.05	5.69	7.75	8.70	8.75	9.16		
	HDD50	3692	877	773	577	241	47	0	0	0	7	125	363	682		
	HDD65	7369	1341	1193	1033	647	337	107	38	50	180	496	801	1146		
	CDD50	2184	0	0	10	46	199	421	561	514	321	99	12	1		
	CDD65	386	0	0	1	2	24	77	133	99	44	6	0	0		
	CDH74	3607	0	0	5	33	257	763	1289	826	389	45	0	0		
	CDH80	913	0	0	0	3	55	219	347	190	92	7	0	0		
(14) Wind	WSAvg	7.6	9.3	9.0	8.6	8.9	7.4	6.3	5.4	5.1	5.9	7.7	8.7	9.1		
(15) Precipitation	PrecAvg	32.0	2.0	2.0	1.8	2.9	3.4	3.1	3.1	3.0	3.2	2.8	2.4	2.2		
	PrecMax	40.6	3.8	5.9	4.5	5.8	7.0	6.9	5.6	5.0	7.4	5.6	5.6	4.3		
	PrecMin	24.6	0.6	0.6	0.7	0.7	1.6	0.9	1.1	0.7	0.4	1.5	0.7	0.7		
	PrecStd	4.0	0.9	1.2	1.0	1.3	1.5	1.5	1.1	1.1	1.5	1.0	1.3	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	54.5	70.2	79.0	85.7	90.0	90.8	90.2	87.6	80.5	65.9	56.8		
		MCWB	50.8	49.5	60.7	61.9	66.9	72.4	75.0	73.3	71.3	68.2	57.0	54.3		
	2%	DB	45.0	45.6	60.8	72.2	81.0	85.8	87.3	84.3	82.2	73.0	60.7	48.2		
		MCWB	42.5	43.1	53.5	58.7	66.4	70.3	72.6	71.4	68.9	62.3	54.1	46.0		
	5%	DB	39.2	40.1	54.0	65.9	76.0	81.9	83.8	81.5	79.0	68.2	55.3	43.5		
		MCWB	36.9	37.3	48.8	55.3	63.8	69.5	71.2	70.8	67.4	61.1	50.1	41.2		
	10%	DB	36.3	36.3	46.7	60.6	72.0	78.8	81.2	78.9	74.7	63.7	52.1	39.3		
		MCWB	34.8	33.9	42.4	51.5	61.2	66.9	69.9	68.4	65.6	57.7	48.3	37.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	51.4	60.5	65.8	72.7	75.6	78.5	77.2	74.9	70.6	59.3	54.4		
		MCDB	51.6	53.1	69.6	74.5	80.8	84.5	85.7	83.5	83.0	78.2	63.6	56.7		
	2%	WB	43.0	43.5	54.4	60.5	69.1	73.5	75.9	74.6	72.2	65.7	55.0	46.5		
		MCDB	44.4	47.0	59.2	68.6	77.1	82.6	83.6	81.9	78.8	70.4	59.4	47.6		
	5%	WB	37.6	37.4	48.9	56.5	66.1	71.4	73.9	72.5	69.9	62.1	51.6	41.8		
		MCDB	39.1	39.6	53.3	64.4	73.0	79.4	80.7	79.0	75.2	66.7	54.6	43.7		
	10%	WB	34.7	34.0	42.9	52.1	63.0	69.0	72.0	70.5	67.4	58.6	47.9	37.6		
		MCDB	36.0	36.0	46.9	58.9	70.0	75.6	78.6	76.0	72.9	63.3	51.7	39.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	14.9	16.1	19.2	21.1	21.5	21.6	20.8	21.6	17.6	14.0	11.0		
		MCDBR	14.8	17.2	24.5	28.9	25.8	24.6	24.5	22.8	25.1	24.0	20.1	15.0		
	5% WB	MCWBR	13.4	14.9	17.9	17.8	14.8	13.3	12.3	12.3	13.4	15.1	15.5	13.3		
		MCWBR	14.2	15.7	23.2	25.8	22.6	21.8	20.1	19.5	20.1	20.4	18.2	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.305	0.325	0.358	0.383	0.410	0.405	0.397	0.368	0.349	0.312	0.282		
	taud		2.389	2.352	2.384	2.341	2.337	2.285	2.329	2.337	2.414	2.442	2.508	2.488		
	Ebn at Noon		263	281	290	287	282	273	273	272	272	262	258	261		
	Edh at Noon		24	30	33	38	39	41	39	37	32	28	22	20		
(44) All-Sky Solar Radiation	RadAvg		405	670	1105	1444	1705	1914	1909	1639	1298	803	471	317		
	RadStd		44	79	114	124	147	108	89	73	122	78	63	33		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.46	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page